Beyond Semantic Confidence: Relation-Algebra-Constrained Geometric Compatibility for 3D Scene Graph Relations

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Abstract

3D Scene Graph predictors can assign high confidence to relation candidates whose predicates are inconsistent with reconstructed object-pair geometry. We formulate compatibility assessment for fixed relation predictions and introduce RelCompat3D, which separates predicate semantics, same-pair geometry, and source confidence. A fitted compatibility layer with family-specific linear heads excludes predictor identity and source scores and learns from linked positive–counterfactual ordering; its scores are projected onto the applicable proximity and vertical relation-algebra orbits. An applicability route re-orders only these families within the source's family slots and preserves selections from support/contact, which lies outside the routing scope. We evaluate exact-label Recall@K jointly with verifier-derived Violation@K across VL-SAT, Open3DSG, and SGFN on a shared 3DSSG target. At $K = 50$, RelCompat3D improves Open3DSG Recall from 0.4043 to 0.4418 while reducing Violation from 0.1387 to 0.0342. Across $K = 10$–50, its point estimates preserve or improve Recall and reduce Violation on all three predictors. Same-route fusion and supervision-matched nonlinear comparisons, factor controls, and family-wise analysis characterize the benefits and limits of the factorized mechanism.

Introduction

3D Scene Graphs turn reconstructed scenes into structured object-relation representations that can support spatial reasoning, embodied AI, scene alignment, visual grounding, and downstream decision making (Wald et al. 2020; Sarkar et al. 2023; Xie, Pagani, and Stricker 2024; Liu et al. 2026). For these uses, relation prediction must be supported by the reconstructed scene, not only plausible at the category or language level. A graph edge such as `standing on`, `next to`, or `higher than` is useful only if it describes the actual subject-object pair in 3D.

We study a failure mode in which relation predictors rank plausible labels that are inconsistent under explicit tests of the actual object-pair geometry. A relation can sound plausible for two object categories while being contradicted by their contact evidence, distance, or vertical arrangement in the reconstructed scene. This mismatch matters more as 3D Scene Graphs become interfaces for language-facing, open-vocabulary, and embodied systems (Koch et al. 2024b; Hou et al. 2025; Zhang et al. 2025; Wang et al. 2025; Saxena and Chiun 2025; Yu et al. 2025; Madhavaram et al. 2026).

The failure is not simply that existing 3DSSG methods ignore geometry. Prior work already uses edge features, point-cloud structure, spatial knowledge, multimodal semantics, and geometric fusion (Zhang et al. 2021; Feng et al. 2023; Wang et al. 2023; Koch et al. 2024a; Chen et al. 2024). The gap is more specific: a source relation score—even when its predictor already uses geometry—is not necessarily an explicit estimate of whether the same subject-object pair satisfies the predicate in 3D. This makes the problem an AI reliability issue, not only a 3DSSG implementation detail: language-facing and embodied systems can inherit a confident symbolic relation that contradicts the underlying world state.

This failure suggests a factor-isolated design. RelCompat3D preserves each relation candidate's subject–object identity, joins its reconstructed geometry, and separates predicate semantics T, same-pair geometry G, and source score Z. A fitted compatibility layer with family-specific linear heads excludes Z and predictor identity; Z enters only during re-ranking. Linked counterfactuals teach the model which geometry supports a predicate, while orbit projection enforces proximity symmetry and invariance under joint endpoint swap and inverse-predicate transformation exactly. An applicability route uses the resulting product only to re-order proximity and vertical candidates within the source's family slots, while support/contact retains its source ordering. Scale-robust fusion, nonlinear, family-conditioning, and rule-filtering comparisons isolate the mechanism. Figure 1 summarizes the framework.

We evaluate support/contact, proximity, and relative-vertical relations because their constraints are identifiable from reconstructed ordered-pair geometry. Open3DSG, VL-SAT, and SGFN provide complementary open-vocabulary and closed-set predictors (Wang et al. 2023; Koch et al. 2024b). The evaluation reports exact-label recall together with verifier-derived violation, uncertainty, coverage, paired confidence intervals, and family composition. We do not extrapolate this cross-predictor evidence across dataset ontologies.

Our contributions are threefold:

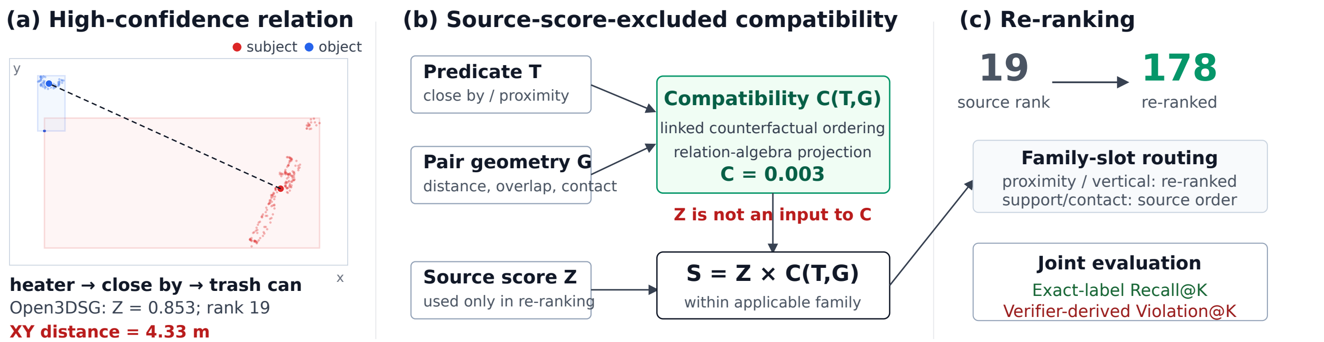

Figure 1: **Failure mechanism and RelCompat3D overview.** A high-scoring Open3DSG relation is contradicted by the actual ordered-pair geometry. RelCompat3D preserves the relation-row identity, isolates predicate semantics T, same-pair geometry G, and source confidence Z, and estimates $C^{\mathrm{alg}}(T, G)$ without Z. The applicability route moves the shown proximity relation from rank 19 to 178 while leaving support/contact selections unchanged. Linked-counterfactual and relation-algebra controls test the factorization; evaluation jointly reports exact-label recall and verifier-derived violation.

1. We formalize the mismatch between relation confidence and instance-level geometric consistency through identity-preserving evaluation and joint exact-label Recall@K–verifier-derived Violation@K.
2. We introduce a source-score-excluded compatibility layer that combines linked counterfactual ordering with exact relation-algebra projection, and an applicability route that preserves selections from families outside the routing scope.
3. We show consistent behavior across three predictors on a shared target and characterize when the reliability layer succeeds or fails through matched controls, uncertainty analysis, and family decomposition.

Across $K = 10$–50 budgets, the applicability route preserves or improves each predictor's Recall point estimate while reducing Violation; at $K = 50$, paired scan-cluster intervals support Violation reduction on all three predictors without a detectable Recall loss. The result is a reliability layer for geometry-checkable relations rather than a replacement relation generator.

Related Work

3D Scene Graph relation prediction. 3D Scene Graphs organize indoor scenes as object-relation structures grounded in 3D space (Armeni et al. 2019). The 3RScan/3DSSG setting established predicate recall for 3D relation prediction (Wald et al. 2020), while incremental and online systems study how scene graphs can be constructed from RGB-D streams or spatial perception stacks (Wu et al. 2021; Hughes, Chang, and Carlone 2022). Recent predictors improve closed-set predicates using edge reasoning, point-cloud features, multi-modal cues, spatial knowledge, self-supervised pretraining, object-centric features, and edge-centric relational reasoning (Zhang et al. 2021; Feng et al. 2023; Wang et al. 2023; Koch et al. 2024a; Heo et al. 2025; Ma et al. 2026). These methods already use geometry; RelCompat3D asks the complementary question of whether a fixed high-scoring predicate is geometrically compatible with the same reconstructed object pair.

Open-vocabulary 3D graphs. Open-vocabulary 3D perception provides queryable object and region features (Peng et al. 2023; Takmaz et al. 2023); graph systems then connect such features to compact scene representations for querying, planning, navigation, online mapping, and language interaction (Gu et al. 2024; Werby et al. 2024; Maggio et al. 2024). Recent 3D scene graph work extends this line toward open-vocabulary objects, open-set relations, VLM features, online graph generation, and functional relations (Koch et al. 2024b; Chen et al. 2024; Hou et al. 2025; Zhang et al. 2025; Wang et al. 2025). This line raises the reliability requirement: when relation edges are language-facing scene representations, plausible text labels must still refer to object pairs that satisfy the corresponding spatial relation.

Geometry-aware relation evidence. A broad set of 3D reasoning methods already uses geometry: edge features, support/contact patterns, affordance geometry, metric distance, topology, and semantic-geometric fusion (Zhang et al. 2021; Feng et al. 2023; Huang et al. 2025; Shao et al. 2025; Xie, Pagani, and Stricker 2024). Scene graph alignment and registration methods further demonstrate that graph structure and geometric consistency matter when scene graphs are reused for matching, alignment, and downstream geometric tasks (Sarkar et al. 2023; Xie, Pagani, and Stricker 2024).

The contemporaneous RelWitness preprint proposes visual-geometric witnesses and witness-consistent decoding for open-vocabulary 3D scene graph generation (Nguyen et al. 2026). Statistical confidence rescoring refines 3D scene graph predictions from multi-view images using semantic masks, neighboring relations, and statistical priors (Yeo, Li, and Lee 2025). These methods modify relation generation or learn source-specific rescoring. RelCompat3D instead operates on fixed candidate outputs and estimates same-pair compatibility with neither predictor identity nor source score as input.

Recent systems sharpen this boundary. SGFormer++ uses a spatial feature adapter inside an incremental predicate predictor (Qi et al. 2026); RelGraphOV prunes geometrically implausible relations while learning an open-vocabulary scene representation (Chen et al. 2026); and Transformation-Aware Decoupling separates predicate reasoning according to how labels transform under yaw-viewpoint changes inside the generator (Sun et al. 2026). In contrast, RelCompat3D leaves each predictor fixed and applies endpoint/predicate algebra to same-pair geometry after prediction, with source score excluded from compatibility.

Neau et al. mine spatial, functional, and relation-algebra constraints and use declarative reasoning to refine fixed image-SGG outputs, reporting a Constraint Violation Rate alongside task metrics (Neau et al. 2026). RelCompat3D differs in estimating continuous compatibility from reconstructed 3D geometry, preserving ordered-pair identity, and enforcing applicable 3D relation-algebra orbits before joint recall–violation evaluation.

Reliability evaluation. Standard R@K asks whether the correct predicate appears near the top, not whether the predicted relation is physically valid in the reconstructed scene. Confidence calibration asks whether model scores reflect empirical correctness likelihood (Guo et al. 2017), and selective prediction makes risk and coverage explicit (Geifman and El-Yaniv 2017). Inspired by this joint-utility view, RelCompat3D standardizes prediction rows, joins same-pair 3D evidence, and reports exact-label recall together with verifier-derived violation. Its compatibility output is a bounded ranking score, not a probability of physical validity.

Method

RelCompat3D assesses the compatibility of fixed relation predictions rather than replacing a 3D Scene Graph generator. It preserves relation-candidate identity, estimates predicate-conditioned geometric compatibility without access to source confidence, and combines the two factors only during re-ranking. Figure 1 summarizes the pipeline.

Problem Setup and Reliability Objective

Let a relation source produce candidate rows

$$r_i = (\mathrm{scan}_i, \mathrm{context}_i, s_i, o_i, p_i, a_i, Z_i), \tag{1}$$

where s_i and o_i are ordered subject and object instance identifiers, p_i is the predicate, a_i its mapped relation family, and Z_i the source relation score. A context is the source-defined ranking unit associated with one scene subgraph; top-K selection is performed independently within each context. The key $(\mathrm{scan}_i, \mathrm{context}_i, s_i, o_i)$ preserves the same object pair across prediction, geometry, and ground-truth joins. We expose predicate/family semantics as $T_i = (p_i, a_i)$ and join raw pair geometry

$$G_i = G(\mathrm{scan}_i, \mathrm{context}_i, s_i, o_i). \tag{2}$$

The measured scope is

$$\mathcal{A} = \{\mathrm{support/contact},\ \mathrm{proximity},\ \mathrm{vertical\ order}\}, \tag{3}$$

whose constraints are identifiable from reconstructed ordered-pair geometry. Rows outside these families are excluded from the measured violation claim. The primary routing scope is narrower: $\mathcal{A}_{\mathrm{alg}} = \{\mathrm{proximity}, \mathrm{vertical\ order}\}$, for which the declared relation algebra is exact. Support/contact is evaluated but passed through because its current evidence lacks a valid family-wide endpoint transform.

Our objective is to estimate a bounded compatibility score

$$C_i = \sigma(h_{a_i}(\Phi(T_i, G_i))) \in [0,1], \qquad Z_i \notin \mathrm{inputs}(C_i), \tag{4}$$

then enforce the applicable relation algebra to obtain C_i^{alg} and form a final ranking score $S_i = F(Z_i, C_i^{\mathrm{alg}})$. The score measures compatibility with a constructed training target: measured-family ground-truth relations are positives, while family-specific high-margin predicate or endpoint interventions generate counterfactual negatives (full rules and thresholds are in the supplement). It is not a probability of physical validity. Predicate-aligned transformations are $T_i \times G_i$ interactions rather than raw geometry. Neither source score nor predictor identity enters C_i, so the fitted compatibility layer can be applied without target-specific refitting to different predictors on the shared target.

Let $L_K(c)$ be the top-K measured-family rows for context c and Y_c its exact-label ground-truth set. We jointly report

$$\begin{aligned} \mathrm{R@K} &= \frac{\sum_c |L_K(c) \cap Y_c|}{\sum_c |Y_c|}, \\ \text{Verifier-V@K} &= \frac{1}{|D_K|} \sum_{(c,i)\in D_K} \mathbf{1}[v_i = \mathrm{violated}]. \end{aligned} \tag{5}$$

where D_K contains selected rows with a supported verifier status $v_i \in \{\mathrm{satisfied}, \mathrm{uncertain}, \mathrm{violated}\}$. Family mapping never relaxes exact predicate matching. Because uncertain rows can lower the all-status ratio without being declared satisfied, the experiments also report decidable-only violation, uncertainty, pessimistic violation, and coverage.

Relation-Algebra-Constrained Compatibility

For each identity-preserving relation row, the raw geometry vector G contains pair distance and scale, centroid and extent differences, oriented-box overlap, and point-level contact/support evidence when observable. The feature map $\Phi = (\phi_T, \phi_G, \phi_\times)$ separates predicate/family indicators, predicate-independent geometry, and explicit predicate–geometry interactions. Source score, rank, source identity, and object-class shortcuts are excluded.

We first define a family logit and bounded compatibility score

$$\begin{aligned} h_a(\Phi(T_i, G_i)) &= w_a^\top \Phi(T_i, G_i), \\ C_i &= \sigma(h_a(\Phi(T_i, G_i)))\,. \end{aligned} \tag{6}$$

For proximity, τ swaps the ordered endpoints and leaves `close by` unchanged. For vertical order, τ swaps the endpoints and maps `higher than`↔`lower than`. These transforms encode

$$\begin{aligned} C_{\mathrm{close}}(s,o) &= C_{\mathrm{close}}(o,s), \\ C_{\mathrm{higher}}(s,o) &= C_{\mathrm{lower}}(o,s). \end{aligned} \tag{7}$$

Support/contact uses the singleton orbit because a family-wide endpoint swap is not valid. Its compatibility model is retained only for unrestricted-fusion diagnostics; the primary route passes this family through unchanged.

Training uses two complementary structural signals. First, orbit-consistent augmentation preserves the predicate when proximity endpoints are swapped, whereas vertical order swaps endpoints and applies the inverse predicate. Second, for every linked positive–counterfactual pair $(i^+, i^-) \in \mathcal{P}$, the logits $\ell = w_a^\top \Phi$ receive a fixed margin penalty:

$$\mathcal{L}_a = \mathcal{L}_{\mathrm{BCE}} + 0.25\, \mathbb{E}_{\mathcal{P}}\left[\log\left(1 + e^{1-(\ell_{i^+} - \ell_{i^-})}\right)\right] + 10^{-4}\|w_a\|_2^2. \tag{8}$$

Numeric features are standardized and missing values are imputed using training-only statistics. Each family uses a small train-only linear model; optimization details and row counts are provided in the supplement.

Finally, we project the learned score onto the declared orbit:

$$C_i^{\mathrm{alg}} = \frac{1}{|\mathcal{O}_i|} \sum_{(T',G') \in \mathcal{O}_i} \sigma(h_{a_i}(\Phi(T', G'))). \tag{9}$$

This makes proximity symmetry and invariance under joint endpoint swap and inverse-predicate transformation exact at inference rather than merely encouraged by a penalty. The pairwise objective still distinguishes each linked positive from its counterfactual; projection does not consume source confidence.

Proposition 1 (Orbit-projection guarantee) *Let a finite relation-algebra group H_a act jointly on predicate and ordered-pair geometry, and let $P_a f(x) = |H_a|^{-1} \sum_{g \in H_a} f(gx)$. Then $P_a f(gx) = P_a f(x)$ for every $g \in H_a$, and $P_a(P_a f) = P_a f$. Consequently, Eq. 9 satisfies proximity symmetry and the corresponding vertical transformation exactly; the singleton support/contact orbit is unchanged.*

Applicability-Routed Re-ranking and Controls

The unrestricted compatibility product preserves source magnitudes through

$$S_i^{\mathrm{alg-prod}} = Z_i C_i^{\mathrm{alg}}. \tag{10}$$

For positive scores, $\log S_i = \log Z_i + \log C_i^{\mathrm{alg}}$; compatibility therefore contributes an additive log-score term without an additional fusion parameter. It is useful when every family has identifiable evidence, but our support/contact diagnostic does not satisfy that condition.

Our primary ranking rule therefore uses family-slot applicability routing. Sort a context by Z and retain its family sequence $(a_1, \ldots, a_n)$. For each family, form

$$Q_a = \begin{cases} \mathrm{sort}_{\downarrow ZC^{\mathrm{alg}}}(a), & a \in \mathcal{A}_{\mathrm{alg}}, \\ \mathrm{sort}_{\downarrow Z}(a), & a = \mathrm{support/contact}. \end{cases} \tag{11}$$

At output position j, take the next unused candidate from Q_{a_j}. This preserves the source-ranked family composition at every K and exactly preserves support/contact selections, while applying compatibility within the proximity and vertical slots. The route introduces no fitted fusion parameter.

Proposition 2 (Family-slot routing guarantee) *For any source ranking and every prefix K, family-slot routing preserves the entire family sequence and the ordered support/contact subsequence. Moreover, among all permutations with that family sequence, it maximizes $\sum_j \omega_j u(e_j)$ for any non-increasing nonnegative position weights ω_j, where $u = ZC^{\mathrm{alg}}$ in routed families and $u = Z$ in pass-through families.*

Proofs are given in the supplement. Proposition 2 formalizes why support/contact Recall and verifier V are identical to the source ranking at every reported K, rather than merely similar empirically.

We also test scale-robust fusion. Let q_i^Z and q_i^C be descending within-context percentile ranks and r_i^Z, r_i^C their one-indexed ranks. Rank-average and fixed-constant Reciprocal Rank Fusion are

$$S_i^{\mathrm{rank}} = \tfrac{1}{2}(q_i^Z + q_i^C), \qquad S_i^{\mathrm{RRF}} = \frac{1}{60 + r_i^Z} + \frac{1}{60 + r_i^C}. \tag{12}$$

They provide scale-robust fusion comparisons. For an apples-to-apples comparison, product, rank-average, RRF, and the supervision-matched nonlinear model are each evaluated through the same family-slot route.

Other comparisons isolate the mechanism. The unrestricted product measures the cost and benefit of applying compatibility to every family. A pooled product replaces the three family models with one $T \times G$ compatibility model fitted on the training split. Compatibility-only removes Z, and an unprojected family product ablates relation-algebra projection. A small nonlinear compatibility model is matched by supervision, features, source-score exclusion, and split; we separately report an exact-label rescorer trained with stronger SGFN-specific supervision.

Linked-counterfactual ordering and wrong-predicate tests ask whether the model uses the intended $T \times G$ interaction. Proximity-swap and vertical-inverse checks test Eq. 9; no blanket endpoint test is applied to support/contact. Identity-preserving joins and the T-only, raw-G-only, additive, shuffled-geometry, and mismatched-pair ablations test score leakage and geometry shortcuts. Together these comparisons test whether the gain follows from factor isolation and algebraic structure rather than an arbitrary fusion choice.

Experiments

Experimental Setup

Datasets and evaluation scope. Open3DSG, VL-SAT, and SGFN provide complementary open-vocabulary and closed-set relation predictors. Together they test cross-predictor robustness under one geometry-identifiable 3DSSG target. All use the same scope: 157 scans, 548 relation contexts, and 3,972 exact-label ground-truth relations from support/contact, proximity, and relative vertical families. Ground-truth membership is used only when computing recall, never to filter or rank candidate relations. The supplement evaluates relative size as a secondary scope extension; a fixed robust-point rule is equally strong within that family, so it is not used as core learned-method evidence.

Table 1: **Joint exact-label Recall (R) and verifier-derived Violation (V; lower is better).** Recall always uses the same 3,972-relation denominator; V uses selected rows with supported verifier status. Routed RelCompat3D is the main ranking rule; rank-average and RRF use the same family-slot route; unrestricted and pooled products are diagnostics. All five budgets are shown, with $K = 50$ used descriptively as a mid-curve reference.

		$K=5$		$K=10$		$K=20$		$K=50$		$K=100$	
Source	Condition	R	V	R	V	R	V	R	V	R	V
Open3DSG	Source score	.0342	.5205	.0987	.3289	.1989	.2099	.4043	.1387	.5111	.1242
Open3DSG	RelCompat3D routed	.0373	.0094	.1135	.0233	.2362	.0313	.4418	.0342	.5692	.0324
Open3DSG	Rank-average (routed)	.0378	.0109	.1108	.0248	.2241	.0353	.4323	.0458	.5408	.0548
Open3DSG	RRF (routed)	.0360	.1899	.1070	.1606	.2163	.1324	.4237	.0948	.5476	.0772
Open3DSG	Unrestricted product	.0999	.0405	.1966	.0452	.3285	.0428	.4655	.0265	.6005	.0330
Open3DSG	Pooled product	.0914	.0525	.1740	.0583	.2925	.0532	.4675	.0510	.6402	.0743
VL-SAT	Source score	.4194	.0029	.6322	.0082	.8074	.0142	.9272	.0268	.9635	.0476
VL-SAT	RelCompat3D routed	.4207	.0015	.6339	.0057	.8082	.0114	.9277	.0197	.9658	.0295
VL-SAT	Rank-average (routed)	.3542	.0015	.5176	.0047	.6949	.0099	.9026	.0162	.9700	.0225
VL-SAT	RRF (routed)	.3678	.0015	.5549	.0051	.7472	.0104	.9174	.0180	.9710	.0247
VL-SAT	Unrestricted product	.4177	.0011	.6332	.0049	.8097	.0110	.9293	.0203	.9688	.0325
VL-SAT	Pooled product	.4172	.0011	.6312	.0062	.8112	.0117	.9300	.0219	.9690	.0387
SGFN	Source score	.3117	.0237	.3975	.0349	.4912	.0322	.7402	.0385	.9235	.0630
SGFN	RelCompat3D routed	.3117	.0237	.3975	.0347	.4914	.0297	.7450	.0263	.9303	.0350
SGFN	Rank-average (routed)	.3109	.0237	.3880	.0347	.4658	.0295	.7022	.0236	.9280	.0259
SGFN	RRF (routed)	.3114	.0237	.3892	.0347	.4703	.0297	.6813	.0253	.8774	.0302
SGFN	Unrestricted product	.3114	.0248	.4061	.0321	.5126	.0274	.7706	.0256	.9418	.0372
SGFN	Pooled product	.3142	.0237	.4079	.0325	.5159	.0278	.7925	.0292	.9413	.0464

Open3DSG uses the official non-averaged model and public preprocessing. Every official context is retained, with unavailable candidate lists treated as empty; coverage and denominator sensitivities are in the supplement.

Conditions and training boundary. We compare the source score and routed RelCompat3D with same-route rank-average, RRF, and supervision-matched nonlinear fusion. Unrestricted and pooled products diagnose routing scope and family conditioning. A separate exact-label rescorer uses stronger SGFN-specific supervision. Controls perturb the predicate, pair identity, geometry stream, endpoint order, source-score factor, or available distance evidence; exact transformations are in the supplement. The compatibility inputs remain limited to T, predicate-independent G, and $T \times G$. Normalization, imputation, and model parameters use only 1,061 training scans; architecture and routing use the 117-scan development split. The 157-scan evaluation split is never used for fitting. All cross-predictor statements concern the shared 3DSSG/3RScan target.

A construct-overlap diagnostic refits train-only models after excluding either the exact verifier input or its broader measurement family; all conditions are reported in the supplement.

Metrics and uncertainty. Recall@K uses exact labels within each relation context. Violation@K is the fraction of selected, geometry-checkable rows classified as violated; uncertain rows are not labeled valid. Decidable-only and pessimistic variants, uncertainty, and coverage test this definition. We report $K \in \{5, 10, 20, 50, 100\}$ and use $K = 50$ only as a descriptive mid-curve reference. Paired 95% intervals use a scan-cluster bootstrap: all contexts from a sampled scan move together, with shared indices across conditions. A paired context-bootstrap sensitivity is provided in the supplement. Within-family and global-top-K family slices expose composition effects that may be hidden by the aggregate metric.

Cross-Predictor Recall–Violation Trade-off

Table 1 and Figure 2 show that the applicability-routed RelCompat3D improves the $K = 50$ trade-off across the three predictors. On Open3DSG and SGFN, paired scan-cluster intervals are strictly positive for Recall and negative for V. On near-ceiling VL-SAT, the Recall interval contains zero while the V interval remains strictly negative, indicating reliability improvement without a detectable Recall loss. Open3DSG exposes the failure most strongly: the source ranks geometrically inconsistent edges at small K, whereas the route changes both recall and violation rather than only pruning edges. Across $K = 10$–50, no routed Recall point estimate is lower than its source counterpart, and V decreases on every predictor. SGFN changes only marginally at $K = 10$, while Open3DSG exhibits the largest gains throughout this range. The $K = 5$ and $K = 100$ columns characterize the low- and high-budget boundaries.

No single comparator dominates across all predictors and operating budgets. Under identical routing (supplement), the matched MLP raises Open3DSG Recall more than the linear product at $K = 50$ (.4670 versus .4418) but also has higher V (.0413 versus .0342); the two are nearly tied on VL-SAT and SGFN. Routed rank-average and RRF often reduce V further at large K but lose more Recall at smaller budgets. The unrestricted product reaches higher aggregate Recall and can lower aggregate V, but changes support/contact selections; this is precisely the family regression hidden by an aggregate score. Pooled product reaches higher Open3DSG Recall at substantially higher V. Rank fusion also incurs predictor-dependent Recall losses. Every routed comparator preserves family composition and support/contact selection exactly; the projected product gives the most consistent Recall–V balance across $K = 10$–50 without a capacity increase. A paired context-level sensitivity in the supplement gives the same directions as the primary scan-cluster analysis.

Structural and Factor Diagnostics

Table 2: **Ablations and counterfactual controls.** R is Recall and V is Violation (lower is better). Every condition uses the same family-slot route and source candidates as Table 1; support/contact remains a source-order pass-through.

Source	Condition	R50	V50	R100	V100
Open3DSG	RelCompat3D routed	.4418	.0342	.5692	.0324
Open3DSG	Wrong predicate	.4265	.2200	.5347	.2098
Open3DSG	Wrong-pair G	.3852	.0848	.4932	.0835
Open3DSG	Shuffled G	.3844	.1293	.4869	.1243
Open3DSG	Label-fixed swap	.4267	.2200	.5373	.2098
Open3DSG	Distance only	.5116	.0824	.6322	.0955
Open3DSG	Compatibility only	.4207	.0342	.5677	.0324
VL-SAT	RelCompat3D routed	.9277	.0197	.9658	.0295
VL-SAT	Wrong predicate	.9043	.0498	.9481	.0796
VL-SAT	Wrong-pair G	.9053	.0261	.9496	.0473
VL-SAT	Shuffled G	.8917	.0306	.9411	.0560
VL-SAT	Label-fixed swap	.9043	.0498	.9481	.0796
VL-SAT	Distance only	.8190	.0534	.8980	.0809
VL-SAT	Compatibility only	.6765	.0161	.8409	.0203
SGFN	RelCompat3D routed	.7450	.0263	.9303	.0350
SGFN	Wrong predicate	.7155	.0943	.8998	.1299
SGFN	Wrong-pair G	.7158	.0399	.8799	.0665
SGFN	Shuffled G	.7062	.0474	.8678	.0836
SGFN	Label-fixed swap	.7155	.0943	.9003	.1299
SGFN	Distance only	.6319	.1000	.8406	.1266
SGFN	Compatibility only	.5279	.0232	.7072	.0230

On the development split, linked positives outrank their counterfactuals in 99.23% of pairs, and correct vertical predicates outrank their inverses in 97.40% of cases. Projection makes proximity-swap and vertical-inverse errors exactly zero on development and across every evaluation row. These results verify the implemented algebra and support the intended counterfactual ordering.

The unprojected product has nearly identical aggregate Recall, while projection removes the applicable algebra errors exactly. The factor ablation reports T-only, true raw-G-only, additive, and interaction conditions in the supplement. A supervision-matched nonlinear compatibility model does not dominate the linear structured model under identical routing: on Open3DSG it gains Recall but increases V, while differences are small on VL-SAT and SGFN. The stronger SGFN-specific exact-label rescorer improves its source endpoint but does not dominate when transferred without target-specific refitting to the other predictors (supplement).

Table 2 applies every perturbation through the same route as the main method. Wrong-pair and shuffled geometry reduce Recall and raise V, showing that the gain depends on the identity-preserving join; wrong predicates and label-fixed swaps sharply increase V. For binary vertical predicates, retaining the label after swapping endpoints induces the same sign contradiction as predicate inversion; proximity is swap-invariant, explaining the two controls' near-identical aggregate values. Distance alone raises Open3DSG Recall but more than doubles V relative to RelCompat3D, and is worse on both metrics for VL-SAT and SGFN. Removing the source score inside routed families can retain low V but loses substantial Recall, especially on VL-SAT and SGFN, confirming that geometry does not replace source confidence. The unrestricted mechanism audit is reported in the supplement.

The held-out-primitive diagnostic further separates the model from the verifier implementation. Removing the exact verifier scalar preserves nearly the full $K = 50$ result. Removing every directly related distance or vertical measurement attenuates the effect: Open3DSG still improves both metrics, while VL-SAT and SGFN retain Recall but have near-source V at $K = 50$. At $K = 100$, all three held-out conditions improve both point metrics on every predictor (supplement). Thus the result is not a literal copy of one verifier scalar, although correlated geometric measurements remain important.

Uncertainty sensitivity gives the same aggregate conclusion at $K = 100$: decidable-only and pessimistic V decrease on every predictor. Uncertainty changes slightly in either direction, so the all-status improvement is not explained by uniformly moving failures into uncertainty.

Qualitative and Family Analysis

Figure 3 shows two geometry-identifiable corrections and the support/contact boundary. The route moves the heater–trash-can proximity edge from rank 19 to 178 because the pair centers are 4.33 m apart, and moves `floor higher than curtain` from rank 1 to 430 because the floor center is 1.02 m below the curtain. It leaves the residual `door lying on floor` support candidate at rank 21; the unrestricted product would promote it to rank 10, illustrating why support/contact is not routed by the current evidence model.

Family-wise intervals prevent aggregate overstatement. Relative vertical supplies most V reduction, proximity improves the global selection, and support/contact is exactly unchanged by construction. The method therefore does not claim to solve contact-heavy relations; full within-family and global-top-K composition tables are provided in the supplement.

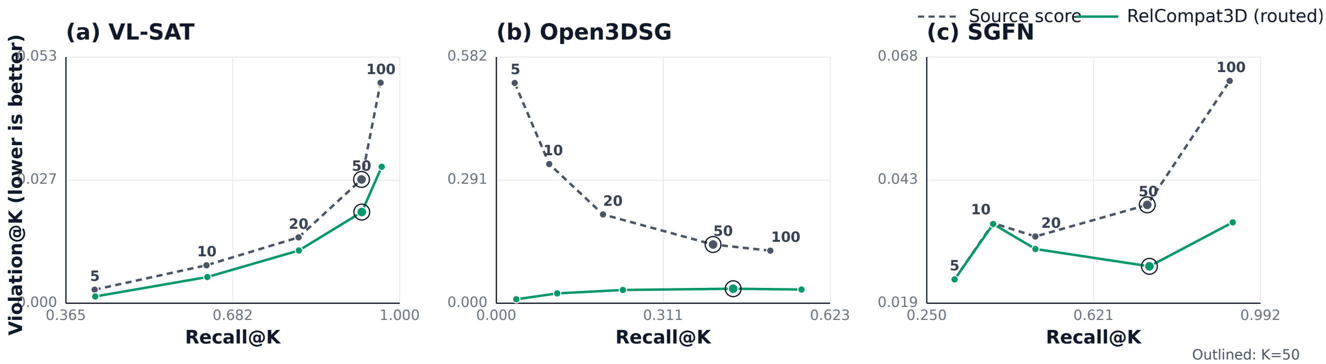

Figure 2: **Recall–violation trajectories.** Each line connects $K \in \{5, 10, 20, 50, 100\}$ for one predictor and ranking rule. Rightward and downward movement is preferred; outlined points denote $K = 50$. Axis ranges differ by predictor.

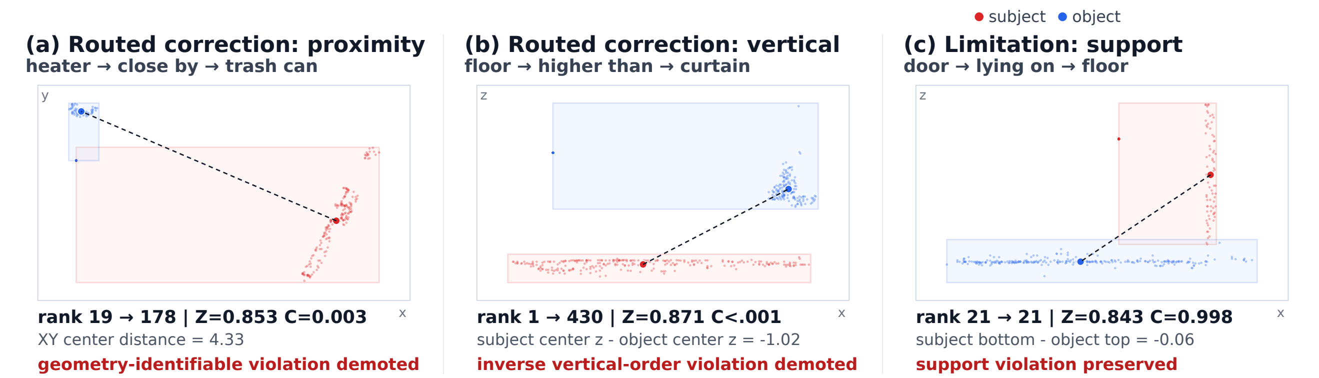

Figure 3: **Geometry-backed re-ranking cases.** Projected views of the ordered-pair point clouds show proximity and relative-vertical violations demoted by the applicability route, plus a residual support/contact violation that remains unchanged. Red denotes the subject and blue the object.

Discussion and Limitations

One fitted compatibility layer with family-specific linear heads is applied without target-specific refitting to three predictors on a shared, geometry-identifiable 3DSSG target, but this does not establish cross-dataset generalization or a universally optimal fusion rule. Rank fusion is stronger at some operating points and pooled compatibility can favor recall over violation. The contribution is a source-score-excluded compatibility layer evaluated across three predictors, together with its joint reliability evaluation, rather than formula superiority.

A ReplicaSSG/FROSS transfer stress test without target-specific refitting improves both metrics at $K = 10$ and $K = 50$ but is nearly inert at $K = 100$ (supplement); it is diagnostic rather than evidence of established dataset-level generalization.

The aggregate effect is not family-uniform. Applicability routing prevents the observed support/contact regression by preserving its source selection, but it also leaves that family's errors uncorrected. Within-family and actual global-top-K slices are reported separately. More broadly, RelCompat3D assumes a geometry-identifiable predicate and known object instances. Functional and reference-frame-dependent predicates remain outside the evaluated scope. The task is relation reliability, not complete open-vocabulary graph generation.

Violation@K is verifier-derived, and some geometry primitives overlap with the constructed compatibility target. Algebra and counterfactual controls test the learned factor, and held-out-primitive refits rule out literal dependence on the exact verifier scalar. Their attenuation after removing the full measurement family nevertheless does not replace an independent physical-validity reference. All three predictors share 3DSSG/3RScan geometry and ontology. Future work should validate the metric independently, improve contact evidence, extend cross-dataset transfer beyond this stress test, and measure downstream decision benefits.

Conclusion

RelCompat3D isolates source confidence from predicate-conditioned object-pair geometry. A linked-counterfactual objective and exact relation-algebra projection turn that separation into a testable compatibility mechanism. Across three predictors on a fixed 3DSSG target, applicability-routed re-ranking improves the Recall–Verifier-V trade-off over $K = 10$–50 while exactly preserving support/contact selections.

References

Armeni, I.; He, Z.-Y.; Gwak, J.; Zamir, A. R.; Fischer, M.; Malik, J.; and Savarese, S. 2019. 3D Scene Graph: A Structure for Unified Semantics, 3D Space, and Camera. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 5664–5673.

Chen, L.; Wang, X.; Lu, J.; Lin, S.; Wang, C.; and He, G. 2024. CLIP-Driven Open-Vocabulary 3D Scene Graph Generation via Cross-Modality Contrastive Learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 27863–27873.

Chen, X.; Hu, J.; Yang, Y.; Zhang, X.; Wang, T.; Bao, H.; Zhang, G.; and Cui, Z. 2026. Beyond Isolated Objects: Relationship-Aware Open-Vocabulary Scene Understanding via 3D Scene Graph Analysis. arXiv:2607.05348.

Feng, M.; Hou, H.; Zhang, L.; Wu, Z.; Guo, Y.; and Mian, A. 2023. 3D Spatial Multimodal Knowledge Accumulation for Scene Graph Prediction in Point Cloud. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 9182–9191.

Geifman, Y.; and El-Yaniv, R. 2017. Selective Classification for Deep Neural Networks. In *Advances in Neural Information Processing Systems*, volume 30.

Gu, Q.; Kuwajerwala, A.; Morin, S.; Jatavallabhula, K. M.; Sen, B.; Agarwal, A.; Rivera, C.; Paul, W.; Ellis, K.; Chellappa, R.; Gan, C.; de Melo, C. M.; Tenenbaum, J. B.; Torralba, A.; Shkurti, F.; and Paull, L. 2024. ConceptGraphs: Open-Vocabulary 3D Scene Graphs for Perception and Planning. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, 5021–5028.

Guo, C.; Pleiss, G.; Sun, Y.; and Weinberger, K. Q. 2017. On Calibration of Modern Neural Networks. In *Proceedings of the 34th International Conference on Machine Learning*, volume 70 of *Proceedings of Machine Learning Research*, 1321–1330.

Heo, K.; Kim, G.; Kim, S.; and Cho, M. 2025. Object-Centric Representation Learning for Enhanced 3D Semantic Scene Graph Prediction. In *Advances in Neural Information Processing Systems*.

Hou, H.-Y.; Lee, C.-Y.; Sonogashira, M.; and Kawanishi, Y. 2025. FROSS: Faster-Than-Real-Time Online 3D Semantic Scene Graph Generation from RGB-D Images. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 28818–28827.

Huang, I.; Bao, Y.; Truong, K.; Zhou, H.; Schmid, C.; Guibas, L.; and Fathi, A. 2025. FirePlace: Geometric Refinements of LLM Common Sense Reasoning for 3D Object Placement. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 13466–13476.

Hughes, N.; Chang, Y.; and Carlone, L. 2022. Hydra: A Real-Time Spatial Perception System for 3D Scene Graph Construction and Optimization. In *Robotics: Science and Systems (RSS)*.

Koch, S.; Hermosilla, P.; Vaskevicius, N.; Colosi, M.; and Ropinski, T. 2024a. SGRec3D: Self-Supervised 3D Scene Graph Learning via Object-Level Scene Reconstruction. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 3404–3414.

Koch, S.; Vaskevicius, N.; Colosi, M.; Hermosilla, P.; and Ropinski, T. 2024b. Open3DSG: Open-Vocabulary 3D Scene Graphs from Point Clouds with Queryable Objects and Open-Set Relationships. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 14183–14193.

Liu, Y.; Mei, H.; Zhan, D.; Zhao, J.; Zhou, D.; Dong, B.; and Yang, X. 2026. View-on-Graph: Zero-Shot 3D Visual Grounding via Vision-Language Reasoning on Scene Graphs. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 40, 7386–7394.

Ma, Y.; Liu, H.; Guo, Y.; Gevers, T.; and Oswald, M. R. 2026. Edge-Centric Relational Reasoning for 3D Scene Graph Prediction. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 40, 7847–7855.

Madhavaram, V.; Sengar, V.; De, A.; and Sharma, C. 2026. VIZOR: Viewpoint-Invariant Zero-Shot Scene Graph Generation for 3D Scene Reasoning. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 8584–8595.

Maggio, D.; Chang, Y.; Hughes, N.; Trang, M.; Griffith, J. D.; Dougherty, C.; Cristofalo, E.; Schmid, L.; and Carlone, L. 2024. Clio: Real-Time Task-Driven Open-Set 3D Scene Graphs. *IEEE Robotics and Automation Letters*, 9(10): 8921–8928.

Neau, M.; Baloch, S.; Suchan, J.; Falomir, Z.; and Bhatt, M. 2026. Visual Commonsense Driven Knowledge Refinements for Scene Graph Generation. *arXiv preprint arXiv:2606.06369*.

Nguyen, M. A.; Tran, Q. H.; Le, B. N.; Pham, T. K.; and Guang, S. Y. 2026. RelWitness: Open-Vocabulary 3D Scene Graph Generation with Visual-Geometric Relation Witnesses. arXiv:2605.20823.

Peng, S.; Genova, K.; Jiang, C. M.; Tagliasacchi, A.; Pollefeys, M.; and Funkhouser, T. 2023. OpenScene: 3D Scene Understanding With Open Vocabularies. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 815–824.

Qi, M.; Lv, C.; Fu, Z.; Zhang, X.; and Ma, H. 2026. SGFormer++: Semantic Graph Transformer for Incremental 3D Scene Graph Generation. arXiv:2606.15328.

Sarkar, S. D.; Miksik, O.; Pollefeys, M.; Barath, D.; and Armeni, I. 2023. SGAligner: 3D Scene Alignment with Scene Graphs. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 21927–21937.

Saxena, P.; and Chiun, J. 2025. ZING-3D: Zero-shot Incremental 3D Scene Graphs via Vision-Language Models. arXiv:2510.21069.

Shao, Y.; Zhai, W.; Yang, Y.; Luo, H.; Cao, Y.; and Zha, Z.-J. 2025. GREAT: Geometry-Intention Collaborative Inference for Open-Vocabulary 3D Object Affordance Grounding. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 17326–17336.

Sun, J.; Wang, C.; Liu, Z.; Tian, J.; Yang, M.; Wang, Y.; and Gao, S. 2026. Not All Relations Rotate Alike: Transformation-Aware Decoupling for Viewpoint-Robust 3D Scene Graph Generation. arXiv:2606.27412.

Takmaz, A.; Fedele, E.; Sumner, R. W.; Pollefeys, M.; Tombari, F.; and Engelmann, F. 2023. OpenMask3D: Open-Vocabulary 3D Instance Segmentation. In *Advances in Neural Information Processing Systems*, volume 36, 68367–68390.

Wald, J.; Dhamo, H.; Navab, N.; and Tombari, F. 2020. Learning 3D Semantic Scene Graphs From 3D Indoor Reconstructions. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 3961–3970.

Wang, Z.; Cheng, B.; Zhao, L.; Xu, D.; Tang, Y.; and Sheng, L. 2023. VL-SAT: Visual-Linguistic Semantics Assisted Training for 3D Semantic Scene Graph Prediction in Point Cloud. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 21560–21569.

Wang, Z.; Su, Y.; Li, C.; Wang, D.; Huang, Y.; Li, X.; and Zhao, B. 2025. Open-Vocabulary Octree-Graph for 3D Scene Understanding. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 7037–7047.

Werby, A.; Huang, C.; Büchner, M.; Valada, A.; and Burgard, W. 2024. Hierarchical Open-Vocabulary 3D Scene Graphs for Language-Grounded Robot Navigation. In *Robotics: Science and Systems (RSS)*.

Wu, S.-C.; Wald, J.; Tateno, K.; Navab, N.; and Tombari, F. 2021. SceneGraphFusion: Incremental 3D Scene Graph Prediction From RGB-D Sequences. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 7515–7525.

Xie, Y.; Pagani, A.; and Stricker, D. 2024. SG-PGM: Partial Graph Matching Network with Semantic Geometric Fusion for 3D Scene Graph Alignment and Its Downstream Tasks. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 28401–28411.

Yeo, Q. X.; Li, Y.; and Lee, G. H. 2025. Statistical Confidence Rescoring for Robust 3D Scene Graph Generation from Multi-View Images. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 24999–25008.

Yu, F.; Deng, Q.; Tang, S.; Li, Y.; and Cheng, L. 2025. Open-World 3D Scene Graph Generation for Retrieval-Augmented Reasoning. Accepted by AAAI 2026, arXiv:2511.05894.

Zhang, C.; Delitzas, A.; Wang, F.; Zhang, R.; Ji, X.; Pollefeys, M.; and Engelmann, F. 2025. Open-Vocabulary Functional 3D Scene Graphs for Real-World Indoor Spaces. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 19401–19413.

Zhang, C.; Yu, J.; Song, Y.; and Cai, W. 2021. Exploiting Edge-Oriented Reasoning for 3D Point-Based Scene Graph Analysis. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 9705–9715.